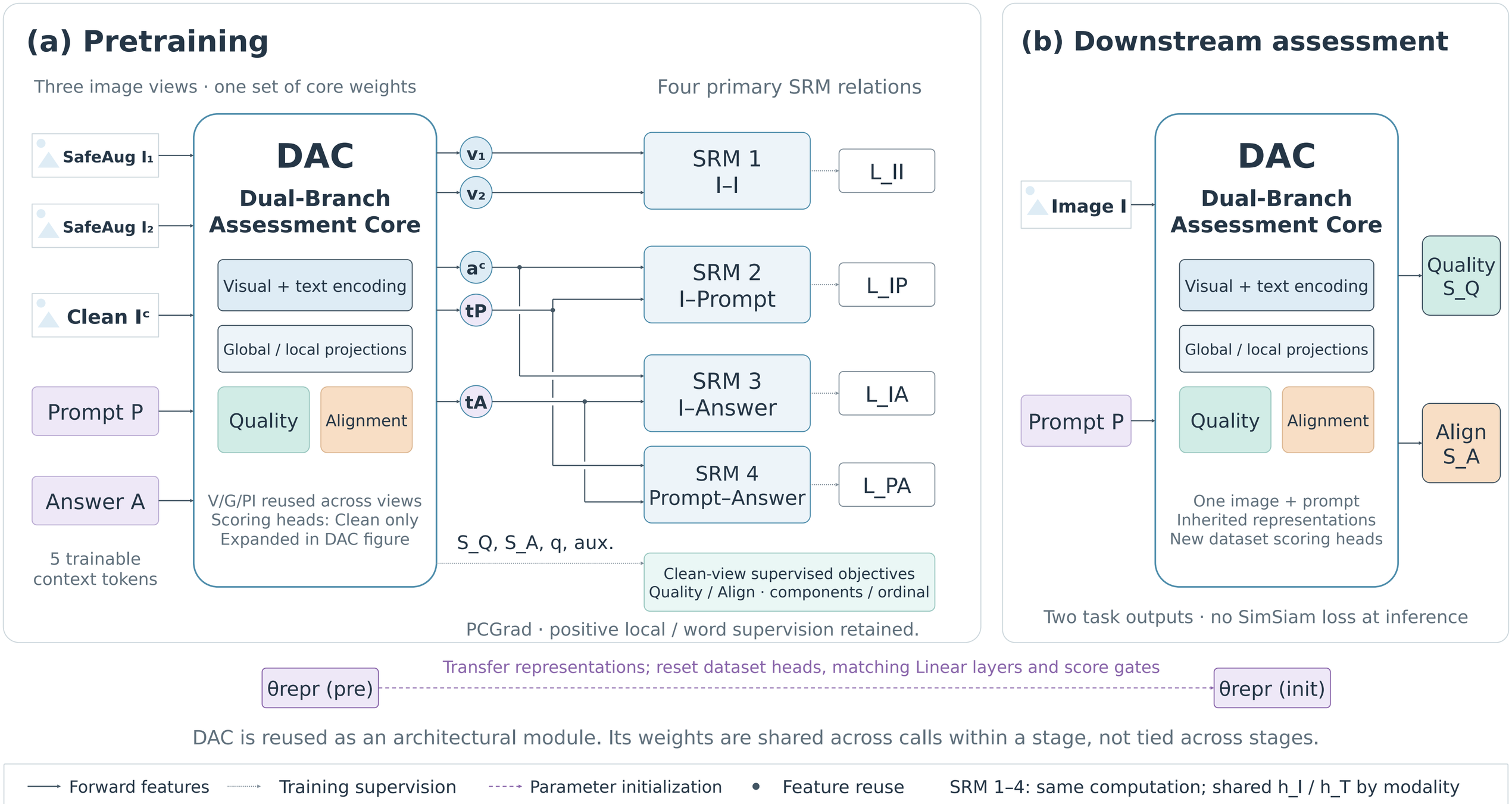

(a) Pretraining
Three image views · one set of core weights
Four primary SRM relations
SafeAug I₁
SafeAug I₂
Clean Iᶜ
Prompt P
Answer A
5 trainable context tokens
DAC
Dual-Branch Assessment Core
Visual + text encoding
Global / local projections
Quality
Alignment
V/G/PI reused across views
Scoring heads: Clean only
Expanded in DAC figure
v₁
v₂
aᶜ
tP
tA
SRM 1 I–I
SRM 2 I–Prompt
SRM 3 I–Answer
SRM 4 Prompt–Answer
L_II
L_IP
L_IA
L_PA
S_Q, S_A, q, aux.
Clean-view supervised objectives
Quality / Align · components / ordinal
PCGrad · positive local / word supervision retained.
(b) Downstream assessment
Image I
Prompt P
DAC
Dual-Branch Assessment Core
Visual + text encoding
Global / local projections
Quality
Alignment
One image + prompt
Inherited representations
New dataset scoring heads
Quality S_Q
Align S_A
Two task outputs · no SimSiam loss at inference
θrepr (pre)
Transfer representations; reset dataset heads, matching Linear layers and score gates
θrepr (init)
DAC is reused as an architectural module. Its weights are shared across calls within a stage, not tied across stages.
Forward features
Training supervision
Parameter initialization
Feature reuse
SRM 1–4: same computation; shared h_I / h_T by modality

DAC · Dual-Branch Assessment Core

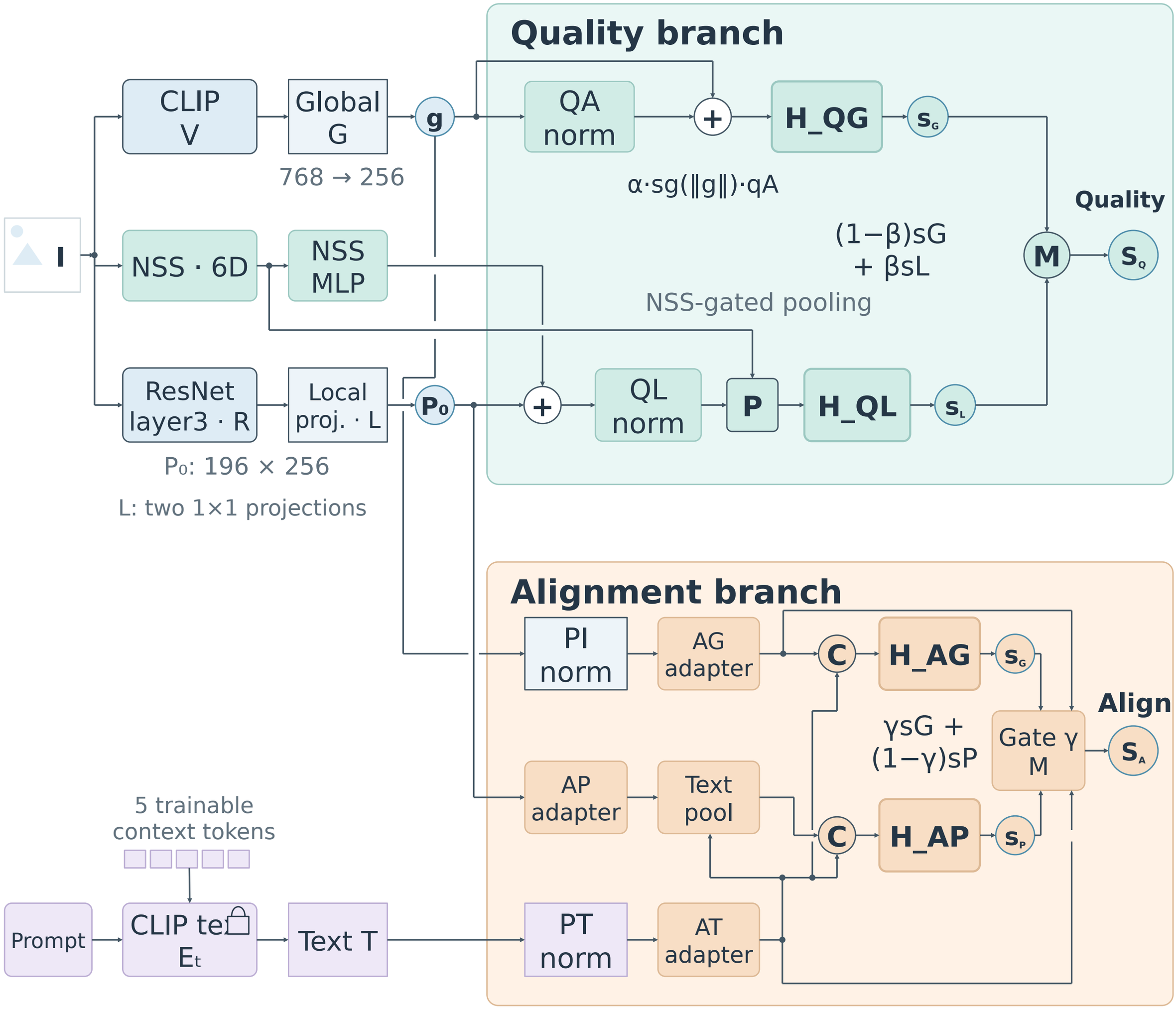


Image-only Quality · text-conditioned Alignment

Training feature taps exported by DAC

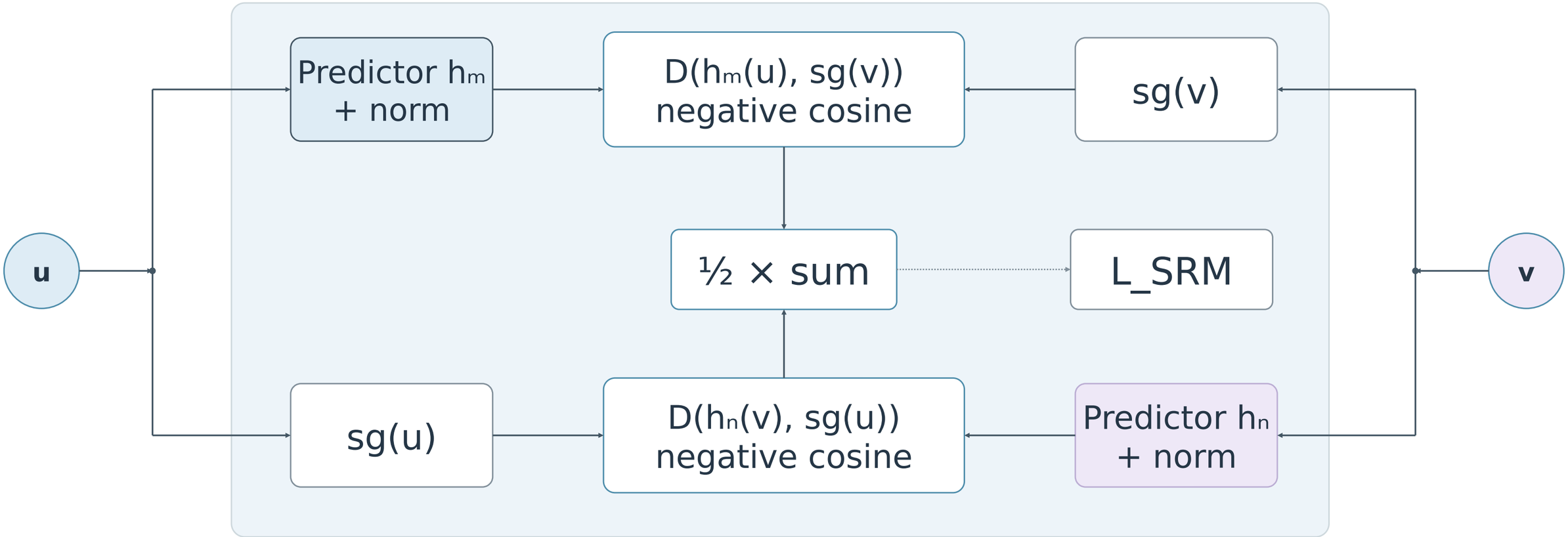
$v = \mathrm{norm}(\mathrm{PI}(g))$; $a^c = \mathrm{norm}(\mathrm{AG}(v^c))$; $q = \mathrm{norm}[(1-\rho)qA + \rho qL]$

$tP = \mathrm{norm}(W_t(\mathrm{AT}(\mathrm{norm}(\mathrm{PT}(T(E_t(P)))))))$; $tA = \mathrm{norm}(\mathrm{PT}(T(E_t(A))))$

Frozen E_t base weights; 5 trainable context tokens. PI/PT are feature projectors, not scoring heads.

SRM · SimSiam Relation Module

Projected feature inputs from DAC · predictors are reused by modality



Four primary relations · two shared modality-specific predictor parameter sets

SRM 1 · I–I	SRM 2 · I–Prompt	SRM 3 · I–Answer	SRM 4 · Prompt–Answer
Inputs: v_1 , v_2	Inputs: a^c , tP	Inputs: a^c , tA	Inputs: tP , tA
Predictors: h_I / h_I	Predictors: h_I / h_T	Predictors: h_I / h_T	Predictors: h_T / h_T

PI/PT remain inside DAC and transfer downstream; h_I/h_T are pretraining-only predictors (256 → 128 → 256).